

Ramya Ajwalia

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EDUCATION

University of Texas at Arlington – Arlington, TX

Aug. 2024 – May. 2026

Master of Science in Computer and Information Science | CGPA: 3.90/4

Gujarat Technological University – India

Jul. 2020 – Jul. 2024

Bachelor of Engineering in Information Technology | CGPA: 9.17/10

EXPERIENCE

AI Product Engineering Intern

Aug. 2025 – Present

Quak Travel – Boston, USA

- Architected FastAPI-based AI recommendation engine reducing inference latency by 20% and improving suggestion accuracy by 15%, saving ~\$20K/year in cloud costs.
- Microservices backend for real-time travel platform; -40% latency, 1.5K+ req/sec throughput.
- Distributed APIs for travel search/itinerary services; -30% inter-service latency, improved scalability.
- Observability for production travel APIs; 99.9%+ uptime, -25% downtime (end users: travellers & booking partners).
- REST APIs + AI inference pipelines (Python, FastAPI, async) powering personalized travel suggestions.
- Backend optimization for consumer-facing travel platform used by end users for trip planning & bookings.

Web Developer Intern

Jan. 2024 – Jun. 2024

Riya SoftWeb Solutions – India

- Fixed 50+ production issues/month in web applications; -25-30% resolution time, +15-20% user satisfaction.
- Managed role-based access control + database updates; +20% data integrity, ensured system stability.
- Created technical documentation for workflows & fixes; -25% recurring issues, improved team efficiency.
- Performed functional + REST API testing for web systems; improved workflow reliability & defect detection.
- Collaborated with developers on debugging & issue resolution for enterprise web applications.
- Supported backend maintenance for business web platforms used for internal operations and client services.

PROJECTS

Driver Drowsiness & Distraction Detection System | [GitHub](#)

Oct. 2025

- Built real-time ADAS using MediaPipe Face Mesh; achieved <33ms/frame latency on CPU-only systems.
- Enabled 5–15% insurance cost reduction (≈ \$50K–\$150K/year) through improved driver safety.
- Developed EAR, MAR, and head pose state machine; improved response time by 0.7s and safety score by 8 points.
- Reduced near-misses and vehicle downtime, improving overall fleet efficiency and operational reliability.

PrismEDU – Full-Stack Learning Management System | [GitHub](#)

Jul. 2025

- Developed React frontend and Node.js/Spring Boot APIs with role-based access and scalable architecture.
- Optimized backend and database queries, reducing API response time by ~25%.
- Tech Stack: React, Node.js, Express, PostgreSQL/MySQL, Python (Flask/FastAPI), AWS, GitHub.
- Performed UI automation using Selenium and REST API testing to validate functionality and integrations.

TECHNICAL SKILLS

- **Programming:** C, C++, Python, Java, JavaScript, TypeScript, SQL, OOP, Data Structures & Algorithms
- **Systems & OS:** Linux, Bash, Process & Memory Management, Multithreading, Concurrency
- **Networking:** TCP/IP, HTTP, Sockets, Distributed Systems
- **Cloud & Backend:** AWS (EC2, S3, RDS), Microservices, Event-driven Systems, Fault Tolerance
- **Testing & QA:** Selenium, REST API Testing, Functional/Integration Testing, Debugging
- **Performance & Debugging:** Logging, Profiling, Exception Handling, Root Cause Analysis
- **DevOps & Tools:** Git, CI/CD basics, Unit Testing
- **Coursework:** Database Administration, Data Analysis & Algorithms, Data Modelling Techniques, Software Engineering, Machine Learning, Cloud Computing & Big Data, Artificial Intelligence, Software Testing and Quality Assurance, Advance Topics in Software Engineering, Data Mining, Computer Vision, Computer Networks